Technische Hochschule Ingolstadt

Pre-Crash Severity Prediction & Orientation Control for Injury Reduction in Unavoidable Collisions.

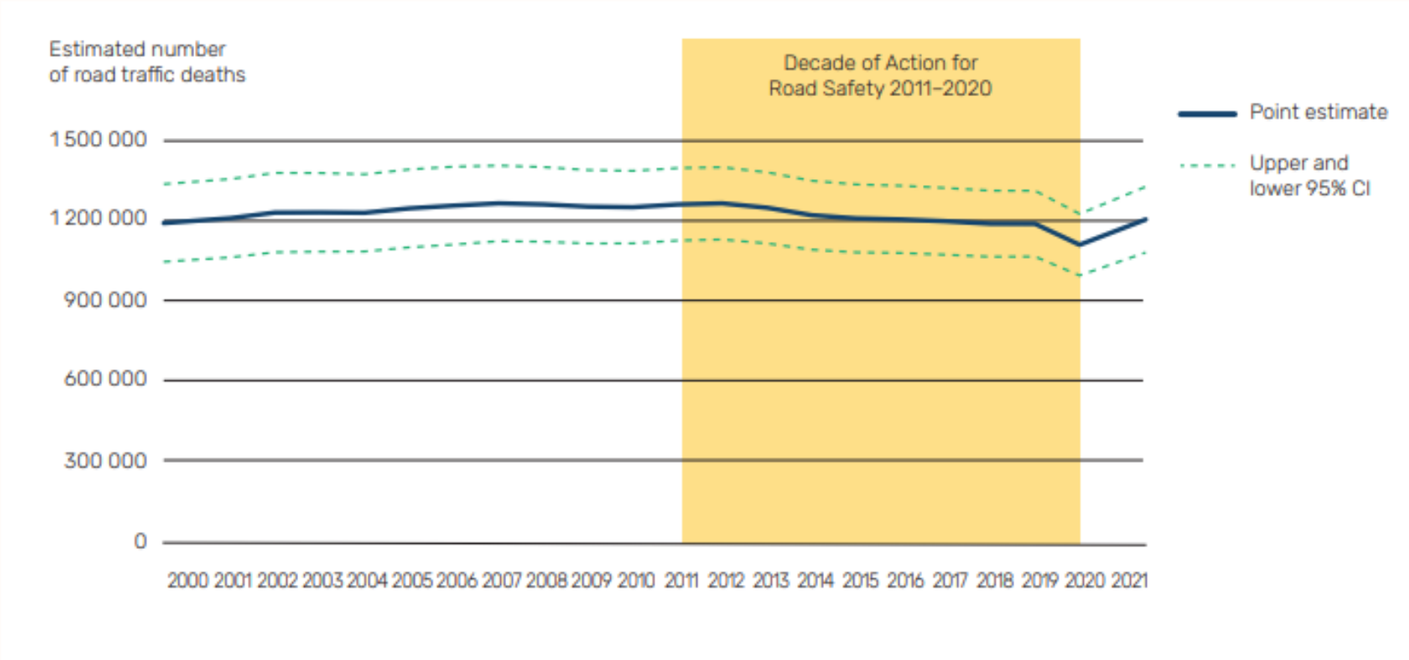
Master Thesis

Author: Aniket Kakde

CARISSMA

The Global Challenge: Road Traffic Fatalities

- According to WHO's Global Status Report on Road Safety 2023, road traffic crashes cause about **1.19 million deaths per year**.
- They are a leading cause of global mortality, especially for people aged **5–29**.
- **Side-impact** crashes have a disproportionately high injury risk due to **limited lateral deformation space**.
- This research investigates **severity-aware control** for **unavoidable** collisions.
- The key idea is to use **precise maneuver** to change **crash geometry** and **reduce injury severity**.



Problem Statement: Severity-Unaware Systems

Current pre-crash systems prioritize collision avoidance through maximum braking (AEB), ignoring biomechanical factors. Existing systems operate in distinct layers:

Pre-Crash Layer

Focuses solely on collision avoidance, and in case the crash is eminent, maximum braking (AEB) is deployed.

In-Crash Layer

Manages energy transfer with restraints and airbags once impact occurs, considering biomechanical factors.

The fundamental gap: Pre-crash systems don't account for injury severity parameters managed by in-crash systems. This thesis bridges that gap by developing severity-aware trajectory planning for the unavoidable collision domain.

Research Questions & Contributions

1. Identify pre-crash kinematic quantities suitable for real-time injury severity assessment.

2. Synthesize Delta-V, impact region, and PDOF into a unified Combined Severity Score $S(\Delta V, x_rel, \alpha)$.

3. Embed severity metric into hierarchical control architecture (APF + NMPC).

4. Validate through simulation: compare risk-aware controller vs. AEB baseline for injury reduction in unavoidable scenarios.

Key Contributions: Unified injury severity metric, Injury severity aware safety function and Hybrid APF-MPC architecture.

Risk-Based Injury Severity Model

Translating physical crash parameters into a quantifiable mathematical objective for injury minimization.

Collision Intensity (Delta-V)

Primary proxy for kinetic energy, predicts MAIS 3+ injury probability.

Structural Stiffness (Impact Region)

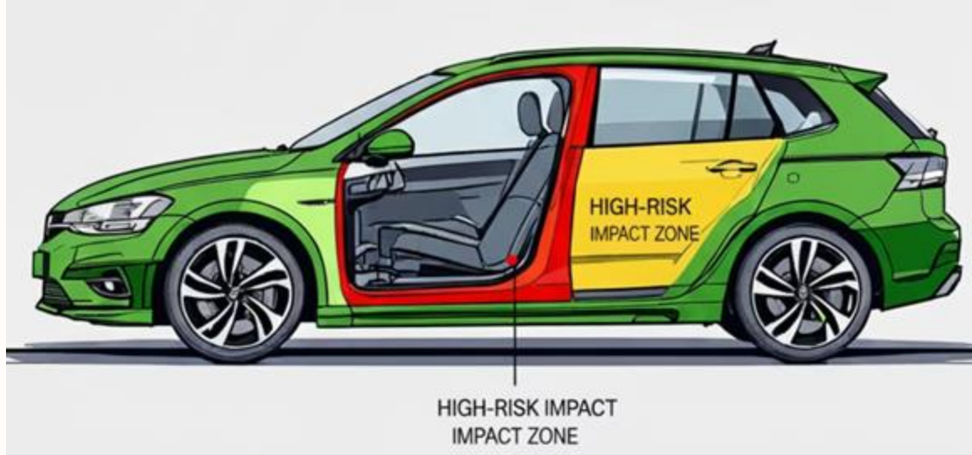
Accounts for heterogeneous vehicle structure; quantifies regions on vehicle's side profile based on injury risk.

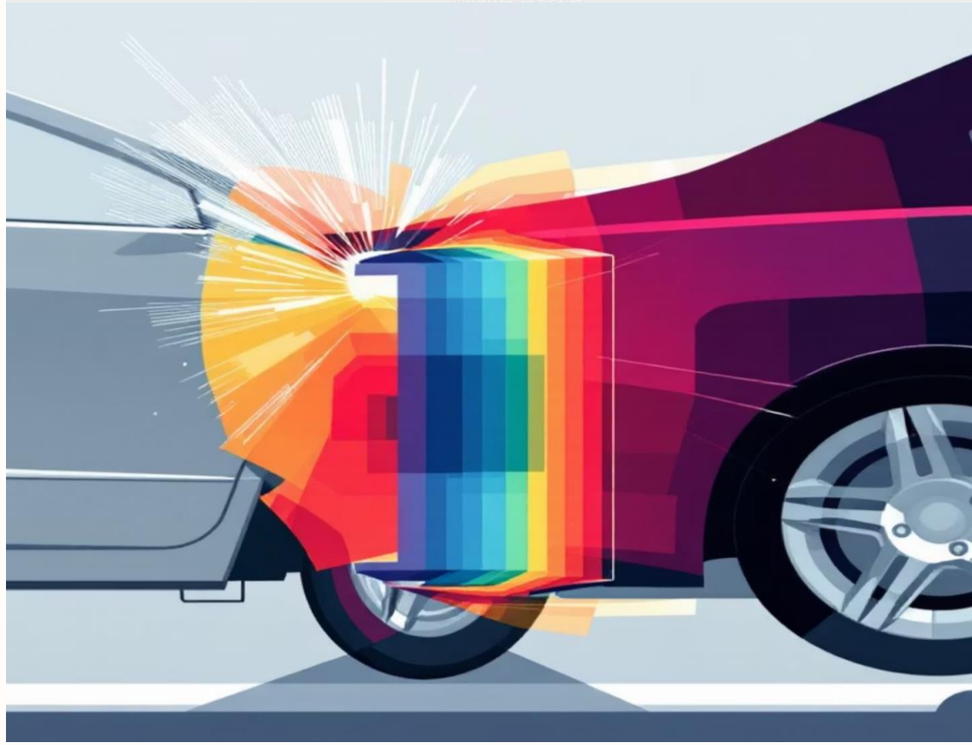
Geometric Orientation (PDOF)

Describes impact angle, with oblique impacts (10 o'clock) increasing pelvic injury risk by up to 1.5x vs. pure lateral (9 o'clock).

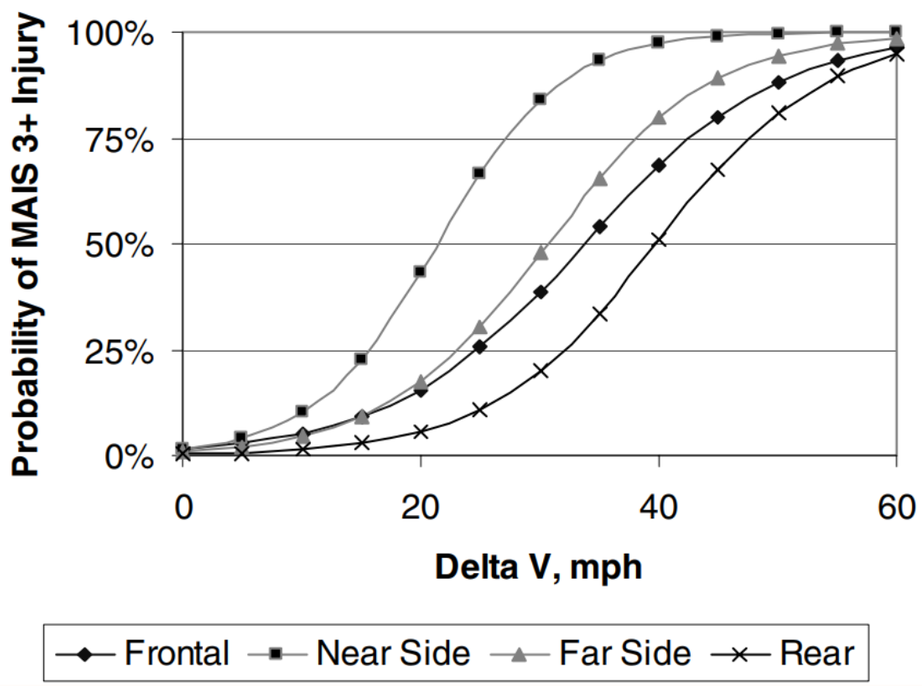
Combined Severity Score (S):

$$S = \lambda_R(Xrel) \cdot \lambda_A(\alpha) \cdot P(\Delta V)$$

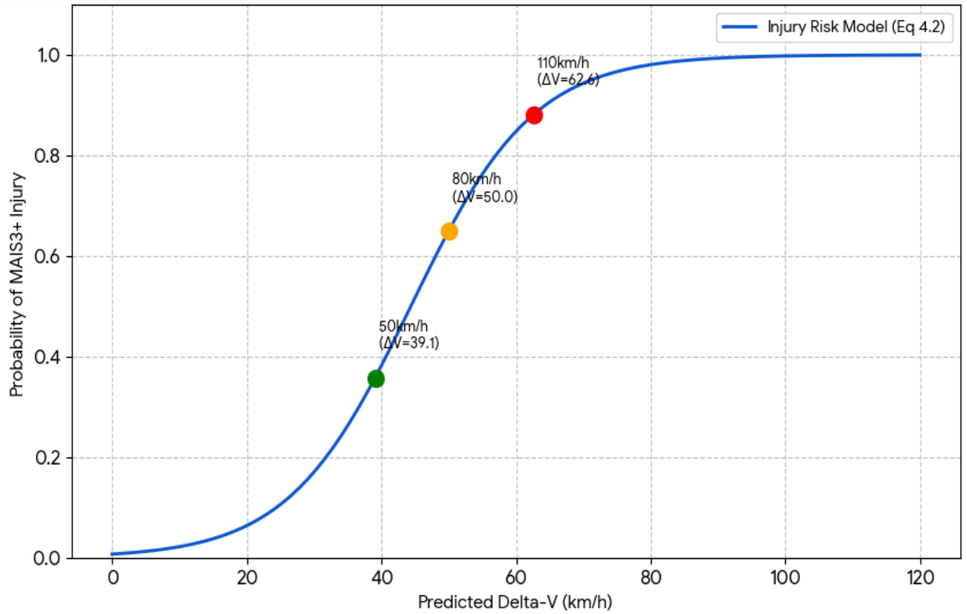




λ_R = Region severity weight.
X_{rel} = Longitudinal location on target vehicle's side profile.
λ_A = Angle severity weight.
α = Angle of Impact.
$P(\Delta V)$ = Probability of MAIS3+ injury based on Delta-V)



"Characteristics of crashes that increase the risk of serious injuries," Proceedings of the 47th Annual Conference (AAAM), (Lisbon, Portugal), pp. 561–576.



Delta-V: The Core of Crash Severity

Defining Delta-V

Change in velocity experienced during impact, perfectly inelastic interaction, gives a conservative estimate.

$$\Delta V_2 = \frac{m_{ego}}{m_{ego} + m_{target}} (V_{ego} - V_{target})$$

Injury Correlation

Delta-V strongly correlates with the probability of MAIS 3+ (serious) injuries.

$$P(\Delta V, \text{Age}) = \frac{1}{1 + \exp\left[-\left(\alpha + \beta_1 \cdot \left|\left|\Delta \mathbf{V_2}\right|\right| + \beta_2 \cdot \text{Age}\right)\right]}$$

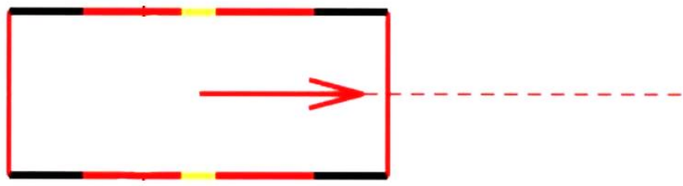
This injury probability model (Augenstein) provides a quantitative link between Delta-V and the likelihood of serious injury. The alpha and beta values are chosen for near-side impacts from the paper.

Risk is negligible below 20 km/h, rises steeply between 30-70 km/h, and saturates near 100% above 80 km/h.

Standardized Metric

Utilized in NHTSA crash test data and the ISO/TS 13405 standard for injury risk assessment.

Impact Region: Structural Stiffness



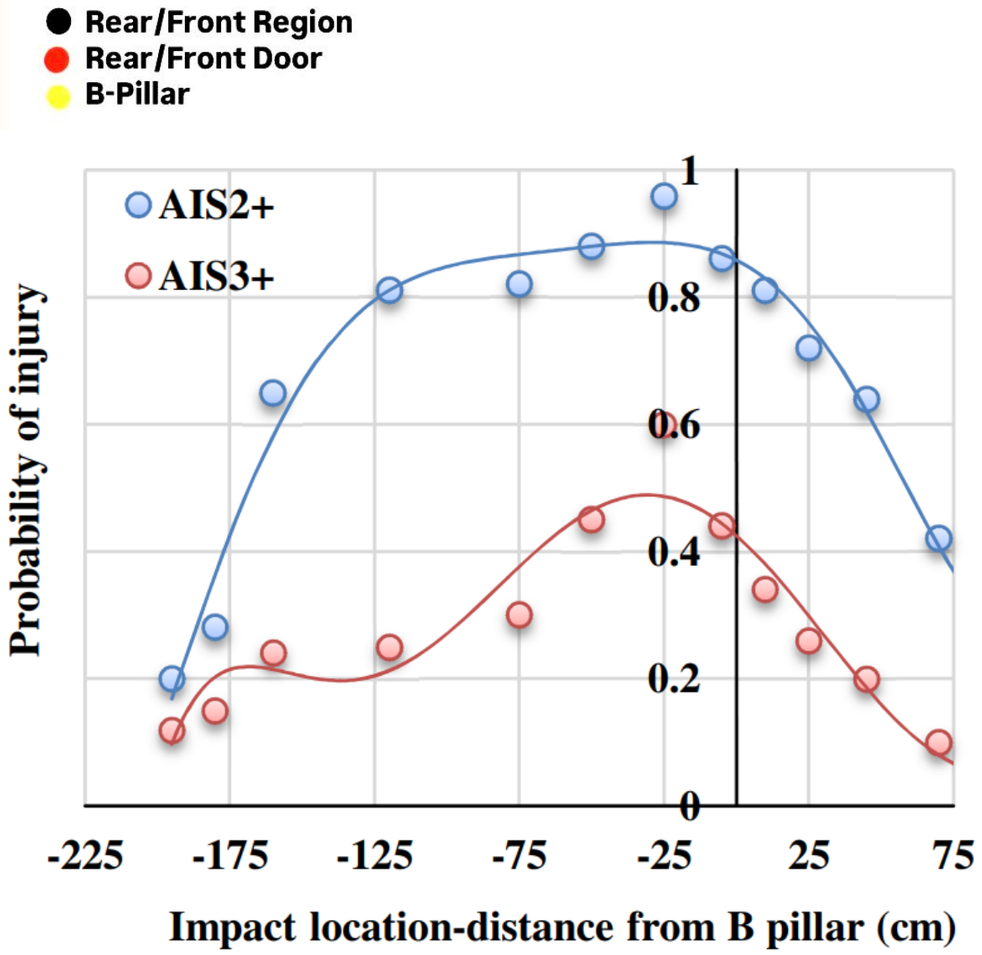
Location Matters

- Evidence: Pal et al. + Ikpe et al. link region stiffness (B-pillar ≈ 140 kN, <10 mm deflection) to the AIS3+ risk hierarchy in NASS-CDS.
- B-pillar (bad): Too stiff → impact impulse transfers as a high acceleration pulse → higher thoracic/pelvic trauma risk.
- Doors (risky): Deform (ride-down lowers peak accel) but can intrude into survival space, raising injury probability.
- Rear/Front regions (best): Sacrificial crumple zones absorb energy before the rigid safety cell, with no immediate cabin intrusion.

Region Weighting Function (λ_R)

- B-Pillar [-0.20, +0.20]m: $\boldsymbol{\lambda_R}$ **= 0.60** (peak stiffness, no deformation).
- Door Panels ±[0.20, 1.40]m: $\boldsymbol{\lambda_R}$ **= 0.45** (compliant, intrusion risk).
- Front and rear regions ±[1.40, 2.30]m: $\boldsymbol{\lambda_R}$ **= 0.20** (crumple zones, energy absorbing).
- Extremities |x| > 2.30m: $\boldsymbol{\lambda_R}$ **= 0.10** (glancing/miss).

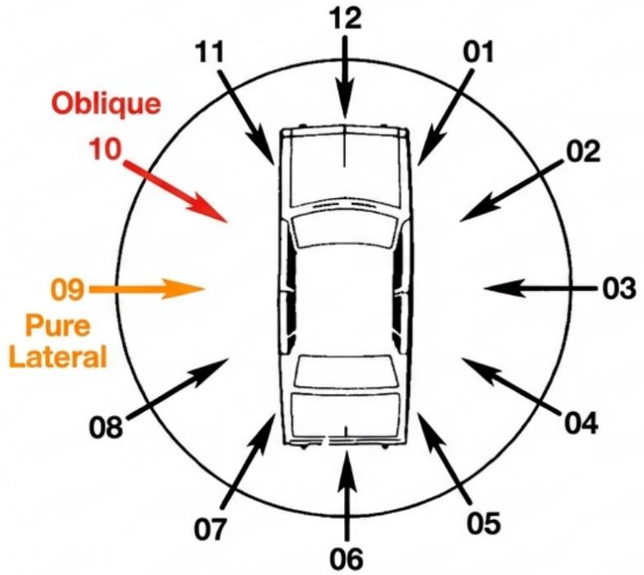
This hierarchy directly correlates with AIS3+ injury probabilities from real-world NASS-CDS data.



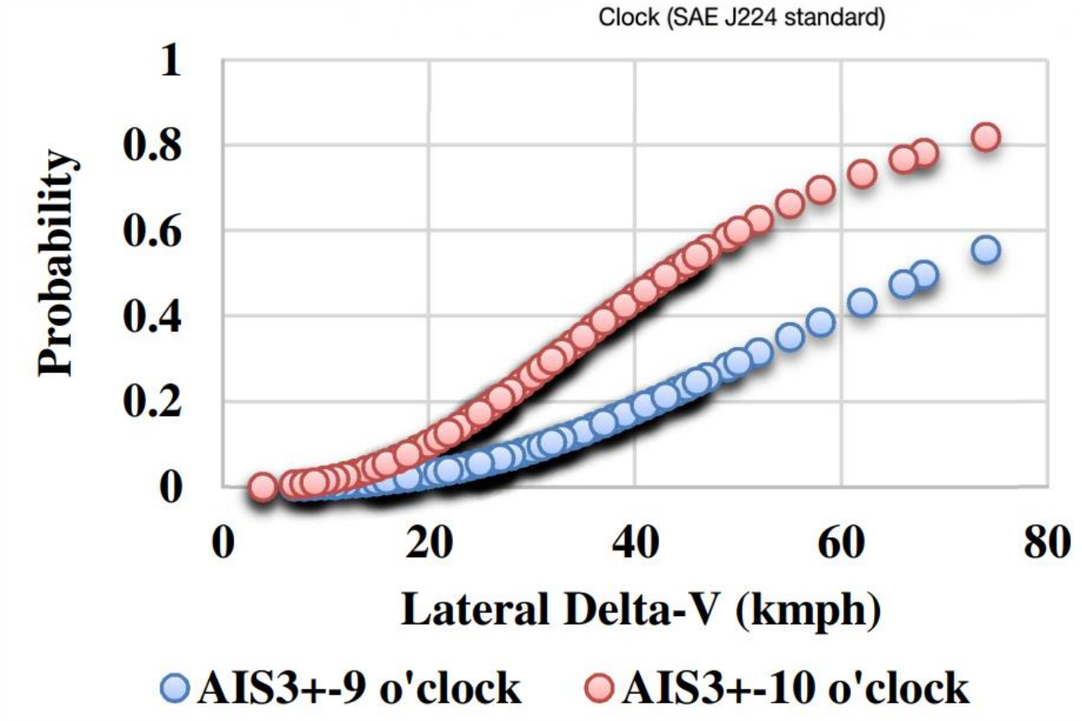
"Real world accident analysis of car-to-car intersection near-side impacts: Focus on pelvis injury," Proceedings of the 26th International Technical Conference on the Enhanced Safety of Vehicles (ESV), 2019. Paper 19-0090

Geometric Orientation: Principal Direction of Force (PDOF)

- **PDOF (α):** impact force direction; regs test **9 o'clock (270°)**, but **10 o'clock (300°)** shows different injury mechanisms.
- **Why 10 o'clock is worse:** longitudinal force component → **airbag bypass / forward rotation** (slides off/misses curtain coverage).
- **Biomechanics:** forward component drives pelvis/knee into interior → **LC-3 pelvic fractures** (unstable, higher mortality vs pure lateral).
- **Dynamic weighting (risk ratio):** $\lambda_A(\alpha, \Delta V) = \frac{P_{pelvis}(\Delta V \alpha)}{P_{pelvis}(\Delta V 270°)}$.
- **Discrete λA bands (used in sim):**
  - **High-risk oblique** $\alpha \in [285°, 315°]$: $\lambda_A = \frac{P(\Delta V|10\ o'clock)}{P(\Delta V|9\ o'clock)}$
  - **Pure lateral** $\alpha \in [255°, 285°]$: $\lambda_A = 1.0$
  - **Else:** $\lambda_A = 0.8$



Clock (SAE J224 standard)



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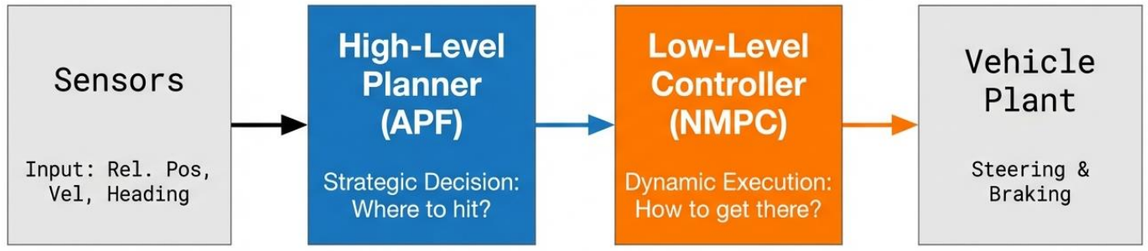
Control Architecture: APF + NMPC



Hierarchical approach separates non-convex planning from dynamic tracking.

The system uses a two-layer hierarchical control architecture for vehicle safety.

Two-Layer Control Hierarchy

High-Level Planner (Artificial Potential Field)

- Generates a risk-minimizing reference trajectory.
- Translates environmental severity into a safe path.
- Adjusts trajectory using attractive and repulsive forces.

Low-Level Controller (Nonlinear Model Predictive Control)

- Precisely executes the APF-generated trajectory.
- Manages braking and steering trade-offs.
- Maintains control within vehicle's physical constraints.

System Parameters

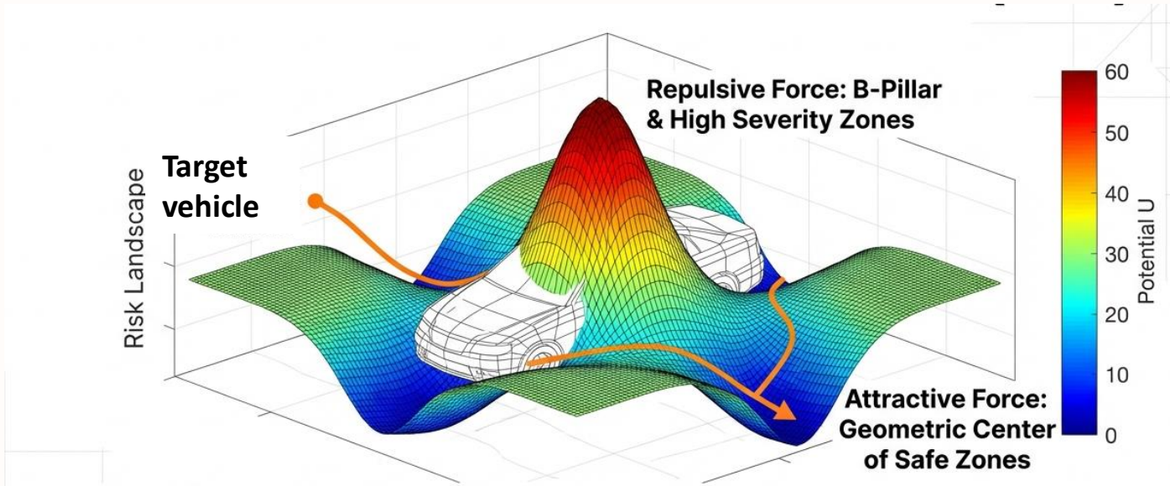
- Prediction Horizon: 25 steps (0.5s)
- Control Horizon: 10 steps (0.2s)
- Execution Frequency: 50 Hz

Artificial Potential Field (APF): Core Concept

The Artificial Potential Field (APF) method guides vehicle control by creating an abstract landscape of forces. Imagine an environment where:

- **Attractive Forces** Pull the ego vehicle towards optimal impact locations, such as deformable crumple zones or away from occupant compartments.
- **Repulsive Forces** Push the ego vehicle away from high-risk areas, like stiff B-pillars or direct occupant contact points, to avoid severe injury.

This transforms the complex injury severity landscape into a navigable potential field. The system then generates a reference trajectory by following the path of least resistance, effectively "rolling downhill" to minimize projected injury severity during an unavoidable collision.



Generates a continuous risk gradient.
Repulsive Force: B-Pillar & High Severity Zones.
Attractive Force: Geometric Center of Safe Zones.

APF: Mathematical Formulation

Longitudinal Attraction (U_att)

This term anchors the ego vehicle to a safe impact zone relative to the leading vehicle.

$$U_{att}(x) = k_{att}(x - x_{des})^2$$

Severity-Based Repulsion (U_rep)

This term creates dynamic barriers around high-risk collision zones, scaling with predicted severity.

$$U_{rep}(x) = \Sigma k_{rep} \quad S_i(\Delta V, \alpha) \quad \exp(-(x - x_{center,i})^2/\sigma^2)$$

Lateral Attraction (U_lat)

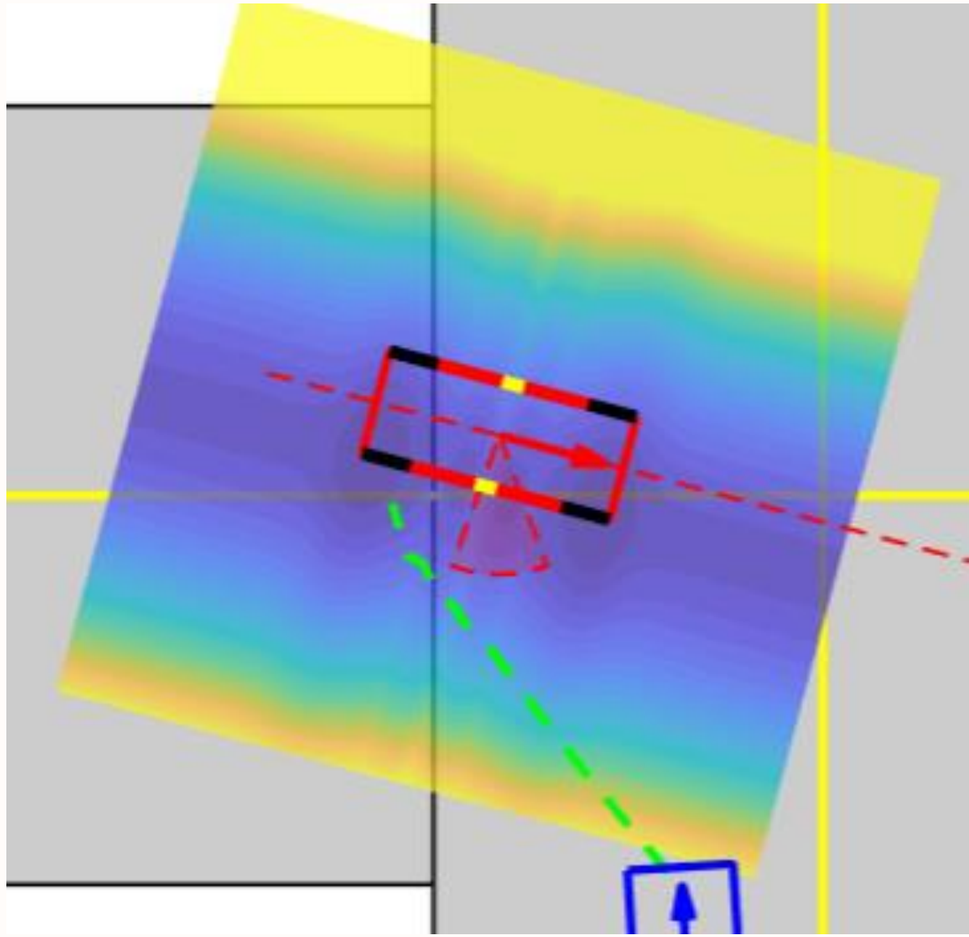
This term aligns the ego vehicle with the target vehicle's lateral position within its lane.

$$U_{lat}(y) = k_y(y - y_{side})^2$$

Total Potential Field

This combines all attractive and repulsive forces to define the overall risk landscape for the vehicle.

$$U(x, y) = U_{att}(x) + U_{rep}(x) + U_{lat}(y)$$



NMPC: Control Strategy & Architecture

NMPC translates high-level guidance into precise vehicle control, optimizing trajectory while respecting dynamic capabilities and physical limitations.

Operational Parameters

- **Prediction Horizon:** Forecasts vehicle states over 0.5 sec (25 steps).
- **Control Horizon:** Optimizes control inputs for 0.2 sec (10 steps) within the prediction window.
- **Control Frequency:** Optimization problem re-solved at 50 Hz for real-time responsiveness.

State and Control Variables

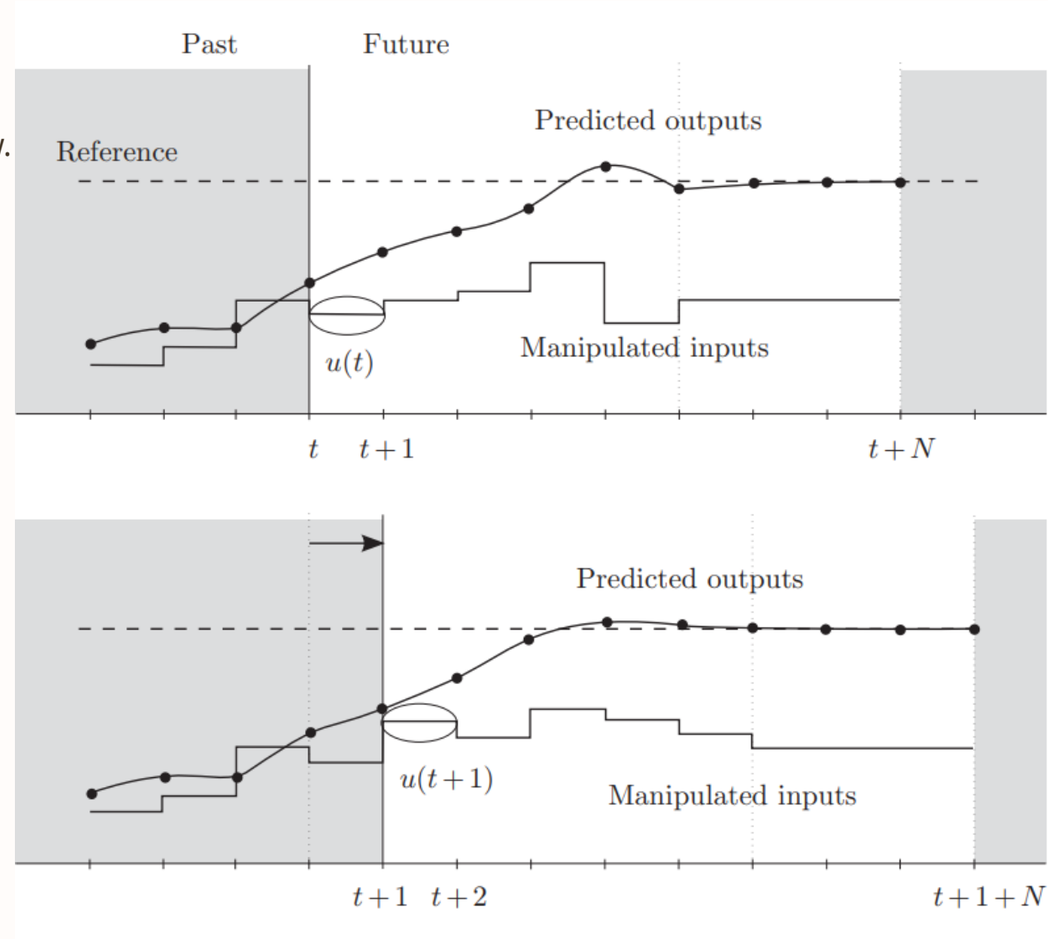
- **States:** Vehicle position, velocity, and yaw rate.
- **Control Inputs:** Steering angle and acceleration (or braking).

Optimization Objectives

- **Reference Path Tracking:** Minimize deviations from desired position, velocity, and heading.
- **Minimize Control Effort:** Ensure smooth and efficient control for stability and comfort.

Constraints

- **Vehicle Dynamics:** Incorporates a non-linear vehicle model and tire characteristics.
- **Friction Limits:** Respects available tire friction to prevent loss of grip.
- **Actuator Saturation:** Limits steering rate and acceleration to vehicle capabilities.



NMPC: Cost Function & Constraints

Cost function:

$$J = \sum_{k=1}^{N_p} \left(w_{pos} \left\| p_k - p_{ref,k} \right\|^2 + w_\psi (\psi_k - \psi_{ref,k})^2 + w_v (v_k - v_{ref,k})^2 \right) + J_{reg}$$

This function minimizes deviations from the desired path and velocity, while also incorporating a regularization term for smooth control.

Cost weights:

- W_{pos} = 100
- W_V = 50
- W_ψ = 10

These weights define the priority given to minimizing errors in position, velocity, and heading, with position tracking being the highest priority.

Regularization term:

$$J_{reg} = \sum_{k=0}^{N_u - 1} \left(0.1 u_{thr}^2 + 0.001 \delta^2 \right) + \left(0.05 \Delta u_{thr}^2 + 0.0005 \Delta \delta^2 \right)$$

This term penalizes large control inputs and rapid changes in control, thereby preventing aggressive vehicle maneuvers.

Friction circle constraint ensures that the combined longitudinal and lateral tire forces remain within the available friction limits, preventing loss of grip.

Vehicle Model and Tire Model

1. Vehicle Model (Nonlinear Single-Track/Bicycle Model)

State Vector:

$$x = [X, Y, \psi, v, r, \beta]^T$$

Where: X, Y = global position; ψ = yaw angle; v = longitudinal velocity; r = yaw rate; β = sideslip angle

Equations of Motion:

Global Kinematics:

$$\dot{X} = v\cos(\psi + \beta); \dot{Y} = v\sin(\psi + \beta); \dot{\psi} = r$$

Longitudinal Dynamics:

$$\dot{v} = a_x$$

Lateral and Yaw Dynamics:

$$\dot{r} = (1/I_k)(lf \quad Fyf \quad \cos\delta - lr \quad Fyr); \dot{\beta} = (Fyf \quad \cos\delta + Fyr)/(m \quad v) - r$$

2. Tire Model (Linear with Saturation)

Slip Angles:

$$\alpha f = \delta - \beta - (lf \quad r)/v; \alpha r = -\beta + (lr \quad r)/v$$

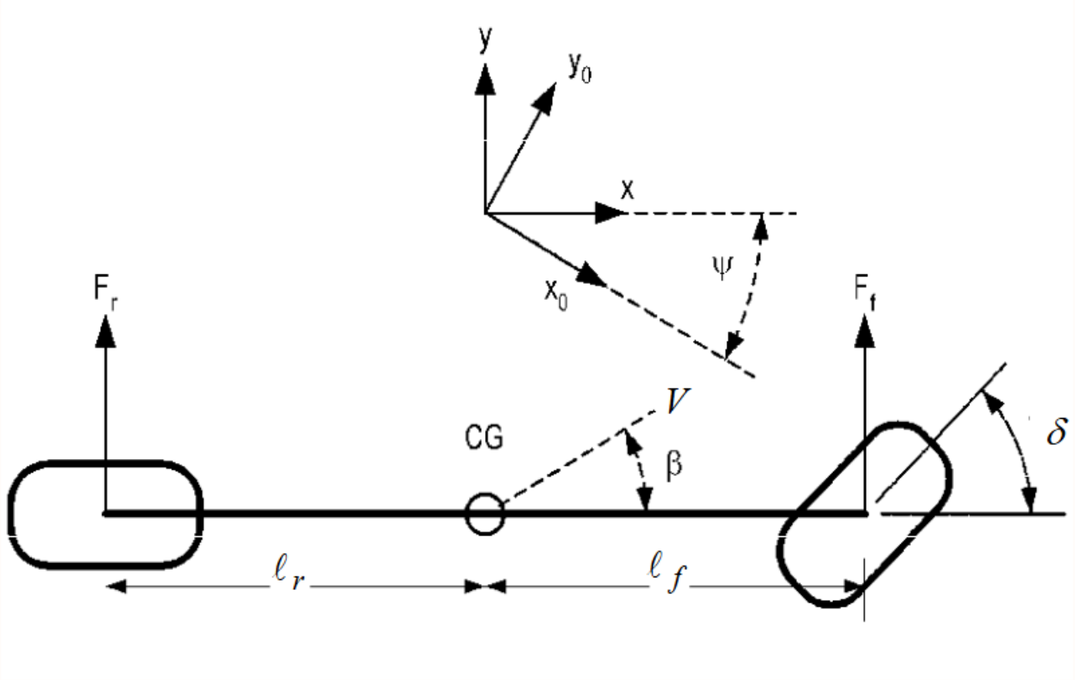
Lateral Force (Linear Region):

$$Fv = C\alpha$$

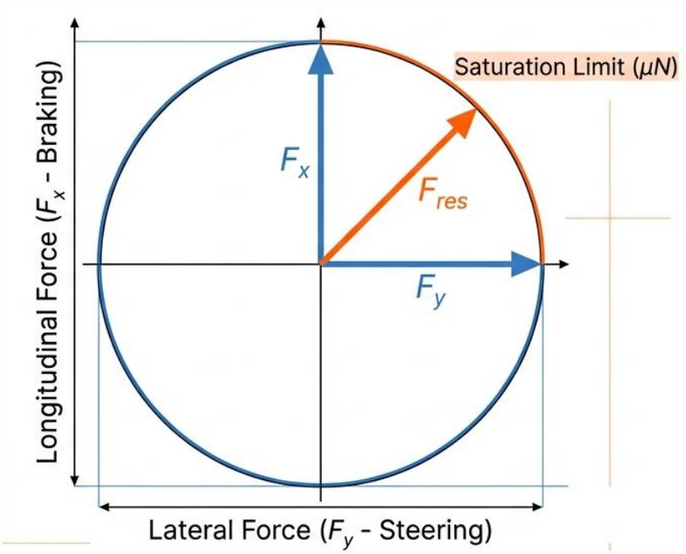
Friction Circle Constraint:

$$Fy, max = \sqrt{[(\mu \quad Fz)^2 - Fx^2]}$$

This constraint couples longitudinal and lateral forces, creating the fundamental trade-off between braking and steering authority.

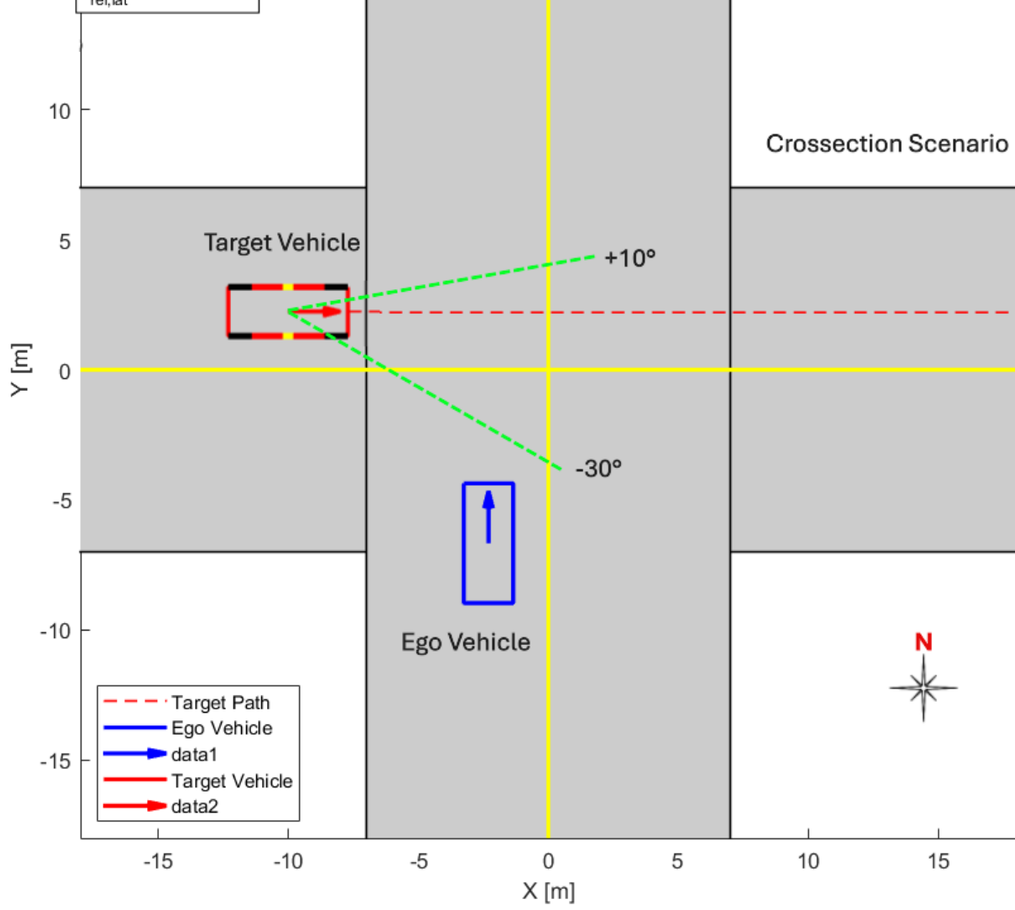


Vehicle Dynamics and Control. New York, NY: Springer, 2nd ed., 2011



Simulation Scenarios & Experimental Design

- **Design:** structured Monte Carlo-style parameter sweep with exhaustive sampling of the operational domain.
- **Vehicle model:** Nonlinear bicycle model; Ego & Target = identical sedans; params in Table.
- **Ego speed levels:** $v_{ego} = \{50, 80, 110\}$ *km/h* these speed ranges are relevant to the Delta-V vs AIS3+ injury risk graph.
- **Target approach angle sweep:** The target vehicle s launched at angles; from $-30°$ to $+10°$ in $5°$ increments, to test robustness of the system.
- **Initial Conditions:** Starting distance and velocity for target vehicle is fixed but varies for ego vehicle according to the iterations.
- **Simulation loop:** For each iteration uncontrolled case, AEB only case and APF+MPC case is simulated and compared for severity score S, Region of impact and Delta-V.



Parameter	Symbol	Value
Total Mass	m	$1600\,\mathrm{kg}$
Yaw Moment of Inertia	I_z	$2500\,\mathrm{kg}\cdot\mathrm{m}^2$
Vehicle Length	L	$4.6\,\mathrm{m}$
Vehicle Width	W	$1.9\,\mathrm{m}$
Distance to Front Axle	l_f	$1.35\,\mathrm{m}$
Distance to Rear Axle	l_r	$1.35\,\mathrm{m}$
Front Cornering Stiffness	$C_{\alpha f}$	$80,000\,\mathrm{N/rad}$
Rear Cornering Stiffness	$C_{\alpha r}$	$100,000\,\mathrm{N/rad}$

Unavoidable Crash Scenario Validation

- **Method:** pre-run Reachability Analysis using idealized point-mass kinematics + friction circle (conservative proof, from Jansson et al.).
- **Stage 1 — braking check:** If stopping is possible → scenario labeled "Avoidable" and discarded.
- **Stage 2 — steering check:** Forward Reachable Set "kinematic funnel" checks for maximum steering angles.
- **Shadow projection test:** Compute conflict window $[t_{in}, t_{out}]$,envelope $(y_{L,max}, y_{R,max})$,then test if target blocks the full funnel.
- **True unavoidable:** Only scenarios that fail both longitudinal + lateral checks proceed to simulation and gives maximum starting distance for ego vehicle as output.
- **Exclude glancing hits:** Apply 0.25 m margin on target extremities → remove bumper/corner "glancing/potentially avoidable" impacts from dataset.

Constraint	Symbol	Limit
Max. Steering Angle	δ_{max}	$\pm 35°$ (0.61 rad)
Max. Braking Deceleration	$a_{x,min}$	$-7.0\,\mathrm{m/s^2}$
Max. Longitudinal Acceleration	$a_{x,max}$	$+6.5\,\mathrm{m/s^2}$
Throttle Slew Rate	$\dot{u}_{thr}$	$25.0\,\mathrm{s^{-1}}$
Steering Rate ($v \leq 60$ kph)	$\dot{\delta}$	$60°/\mathrm{s}$
Steering Rate ($60 < v \leq 100$ kph)	$\dot{\delta}$	$45°/\mathrm{s}$
Steering Rate ($v > 100$ kph)	$\dot{\delta}$	$15°/\mathrm{s}$

Target Angle	Critical Launch Distance (d_{crit}) [m]		
ψ_{tgt} (°)	50 km/h	80 km/h	110 km/h
$-30°$	*Avoidable/Miss*	18.5	24.3
$-25°$	12.1	17.3	22.9
$-20°$	10.9	16.2	21.6
$-15°$	10.4	15.4	20.3
$-10°$	9.5	14.4	19.1
$-5°$	8.8	13.7	18.4
0° (Perpendicular)	7.9	12.7	17.7
$+5°$	7.3	12.0	16.4
$+10°$	6.5	11.2	15.7

"Decision making for collision avoidance systems," SAE Transactions: Journal of Passenger Cars - Electronic and Electrical Systems, vol. 114, pp. 326–335. SAE Technical Paper 2005-01-0422

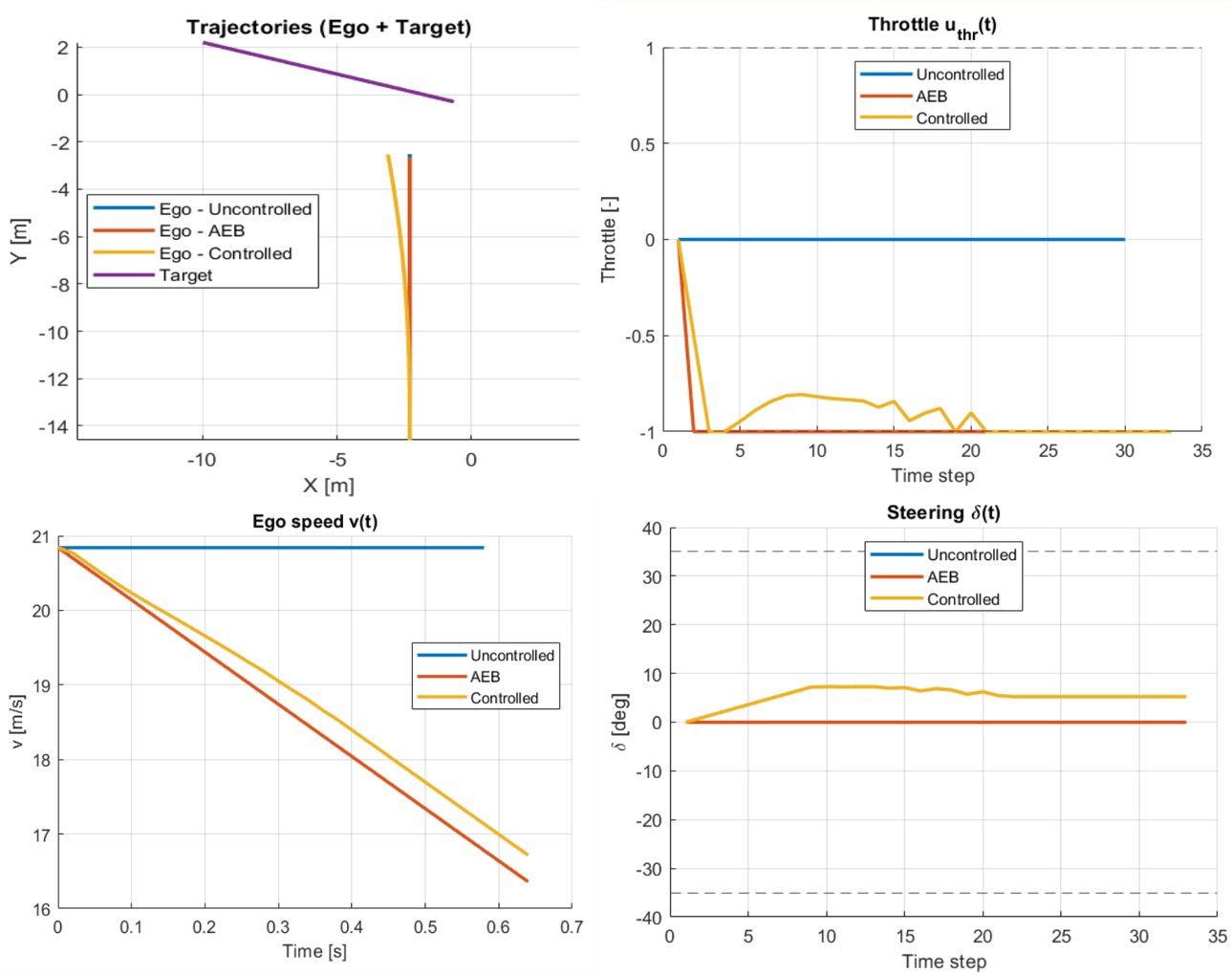
Example Simulation: 75 km/h, -15° Oblique Impact

- **Baseline (no intervention):** impact severity is high with ΔV=10.70 m/s (38.52 km/h) and S=0.526S, impacting front door.

- **AEB-only effect:** mainly reduces crash energy—ΔV falls to **8.04 m/s** (28.94 km/h) and severity drops to **S=0.299** vs **0.526** baseline, impacting back door.

- **APF+MPC combined control:** achieves the lowest severity by further reducing ΔV to **7.77 m/s** (27.98 km/h) and S to **0.122**, i.e., **77%** lower than baseline, impacting comparatively safer region region.

- **APF+MPC vs AEB-only:** despite similar ΔV (7.77 vs 8.04 m/s), severity improves strongly: **0.299** → **0.122** (**59%** reduction).



Angle: -15 deg

Case	Collision	Region	$\Delta V2_{lat}$ [m/s]	$V_{re.lat}$ [m/s]	s
Uncontrolled	YES	front door	10.06	20.13	0.5259
AEB only	YES	back door	8.04	16.07	0.2990
Controlled	YES	rear region	7.77	15.54	0.1217

Simulation Results: Comprehensive Speed & Angle Analysis

Urban Speeds (50 km/h)

APF+MPC demonstrates strong performance in urban scenarios.

- 75-87% injury risk reduction vs Uncontrolled baseline.
- 60-75% injury risk reduction vs AEB-Only baseline.

Rural Speeds (80 km/h)

Consistent injury risk reduction observed across rural conditions.

- 72-80% injury risk reduction vs Uncontrolled baseline.
- 56-65% injury risk reduction vs AEB-Only baseline.
- Critical B-pillar avoidance achieved at -10° target approach angle.

Ang	Dist	Uncontrolled			AEB Only			APF+MPC				
(°)	(m)	Reg	ΔV	S	Reg	ΔV	S	Reg	ΔV	S	% v. Un	% v. AEB
-30	-	SKIPPED (Miss/Glance)										
-25	12.1	F.Reg	6.29	0.09	F.Door	4.45	0.08	**R.Reg**	3.87	0.02	78%	75%
-20	10.9	F.Reg	6.53	0.10	F.Door	4.68	0.09	**Back Dr**	4.26	0.07	30%	22%
-15	10.4	F.Door	6.71	0.23	Back Dr	4.68	0.09	**R.Reg**	4.43	0.03	87%	67%
-10	9.5	F.Door	6.84	0.24	Back Dr	4.84	0.10	**R.Reg**	4.65	0.04	83%	60%
-5	8.8	Back Dr	6.92	0.25	Back Dr	4.83	0.09	**R.Reg**	4.80	0.04	84%	56%
0	7.9	Back Dr	6.95	0.25	R.Reg	4.92	0.04	**R.Reg**	4.96	0.05	80%	-25%
+5	7.3	Back Dr	6.92	0.25	R.Reg	4.90	0.04	**R.Reg**	4.94	0.04	84%	0%
+10	6.5	Back Dr	6.84	0.24	R.Reg	4.91	0.04	**R.Reg**	5.03	0.05	79%	-25%

Ang	Dist	Uncontrolled			AEB Only			APF+MPC				
(°)	(m)	Reg	ΔV	S	Reg	ΔV	S	Reg	ΔV	S	% v. Un	% v. AEB
-30	18.5	F.Reg	9.63	0.26	F.Door	7.75	0.34	**R.Reg**	7.08	0.12	54%	65%
-25	17.3	F.Door	10.07	0.66	F.Door	8.17	0.39	**R.Reg**	7.63	0.14	79%	64%
-20	16.2	F.Door	10.45	0.72	Back Dr	8.47	0.43	**R.Reg**	8.12	0.17	76%	60%
-15	15.4	F.Door	10.74	0.77	Back Dr	8.71	0.46	**R.Reg**	8.46	0.19	75%	59%
-10	14.4	**B-Pillar**	10.95	**1.07**	**Back Dr**	8.88	0.48	**R.Reg**	8.81	0.21	80%	56%
-5	13.7	Back Dr	11.07	0.82	R.Reg	8.98	0.22	**R.Reg**	9.01	0.22	73%	0%
0	12.7	Back Dr	11.12	0.83	R.Reg	9.09	0.23	**R.Reg**	9.14	0.23	72%	0%
+5	12.0	R.Reg	11.07	0.37	R.Reg	9.05	0.22	**R.Reg**	9.20	0.23	38%	-5%
+10	11.2	Back Dr	10.95	0.80	R.Reg	9.02	0.22	**R.Reg**	9.25	0.24	70%	-9%

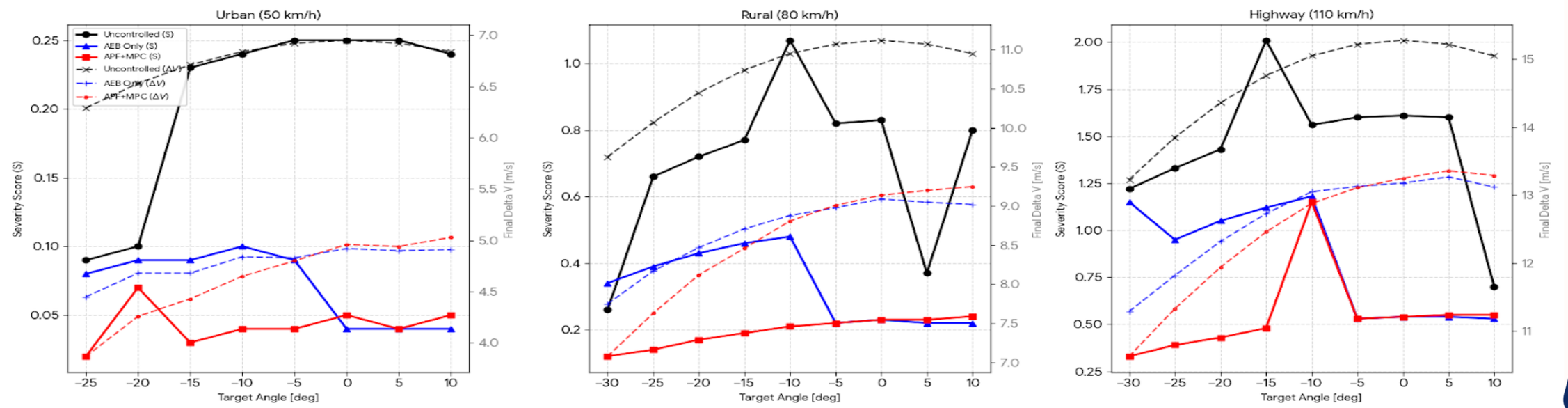
Simulation Results: Comprehensive Speed & Angle Analysis

Highway Speeds (110 km/h)

Benefits diminish at highway speeds due to actuator saturation, yet still effective.

- 40-55% injury risk reduction vs Uncontrolled baseline.
- 30-45% injury risk reduction vs AEB-Only baseline.
- AEB hitting B-Pillar exhibit possible dangerous crash constellations.

Ang	Dist	Uncontrolled			AEB Only			APF+MPC				
(°)	(m)	Reg	ΔV	S	Reg	ΔV	S	Reg	ΔV	S	% v. Un	% v. AEB
-30	24.3	F.Door	13.23	1.22	**B-Pillar**	11.29	**1.15**	**R.Reg**	10.63	0.33	73%	71%
-25	22.9	F.Door	13.85	1.33	Back Dr	11.82	0.95	**R.Reg**	11.33	0.39	71%	59%
-20	21.6	Back Dr	14.36	1.43	Back Dr	12.32	1.05	**R.Reg**	11.94	0.43	70%	59%
-15	20.3	**B-Pillar**	14.76	**2.01**	Back Dr	12.73	1.12	**R.Reg**	12.46	0.48	76%	57%
-10	19.1	Back Dr	15.05	1.56	Back Dr	13.05	1.18	**Back Dr**	12.88	1.15	26%	3%
-5	18.4	Back Dr	15.22	1.60	R.Reg	13.13	0.53	**R.Reg**	13.11	0.53	67%	0%
0	17.7	Back Dr	15.28	1.61	R.Reg	13.18	0.54	**R.Reg**	13.25	0.54	66%	0%
+5	16.4	Back Dr	15.22	1.60	R.Reg	13.27	0.54	**R.Reg**	13.36	0.55	66%	-2%
+10	15.7	R.Reg	15.05	0.70	R.Reg	13.12	0.53	**R.Reg**	13.29	0.55	21%	-4%



Conclusions & Future Outlook

Main Findings

The severity-aware controller demonstrates superior performance, successfully shifting the impact point from the B-pillar to the rear region, achieving up to 56% injury risk reduction in best cases. While physical saturation limits steering authority at highway speeds, meaningful injury risk reduction is still maintained in oblique collision scenarios.

Future Work

High-fidelity crash intensity estimation using neural networks.

Omnidirectional risk mapping (360° severity landscape).

Robustness to uncertainty in sensing and friction estimation.

Herzlichen Dank für Ihre Aufmerksamkeit!